

OSM / Opto-PING Rung 0K

External Reproducibility Handoff Report v0.1

Executive verdict

Rung 0K is a handoff pass, not a true external-reproduction pass. The frozen Rung 0J package was extracted into a fresh clean-room directory and rerun without modifying the locked model equations, parameter grids, perturbation order, seed policy, alternative models, pass/fail criteria, or reporting template. The internal clean-room replay reproduced the locked metrics. A true external reproduction still requires another machine or independent reproducer to run this package unchanged.

Locked specification

Locked spec SHA-256:

d93f6146e4e77ffeffbdd5c133250fd6d474317d8af6c34c3354dd6b28756282

Clean-room replay results

Criterion	Result	Status
n trials	60	PASS
K within 15%	1.000	PASS
K within 10%	1.000	PASS
Median K error	0.028175	PASS
Directional eta/zeta <=20%	1.000	PASS
Full reciprocal CV fold wins	0.872222	PASS

CV winners

```
{
  "full_reciprocal": 314,
  "replay_only_basis": 20,
  "zeta_only": 14,
  "eta_only": 9,
  "generic_hidden_oscillator": 3
}
```

Frozen elements

- model equations
- parameter grids
- perturbation order
- K-lock criteria
- failure thresholds
- random-seed policy
- alternative models
- reporting template

Boundary

This is synthetic identifiability work. It does not prove OSM, microtubular memory, LTP, dendritic spine growth, quantum memory, or any human EEG mechanism. It only tests whether a locked synthetic model can reproduce its primary composite-coupling result.

Status taxonomy

Status	Meaning
HANDOFF_READY	Locked files and instructions are packaged.
INTERNAL_CLEANROOM_PASS	Same locked package reran in a fresh directory.
EXTERNAL_PASS	Independent person/machine reran package unchanged and passed.
EXTERNAL_FAIL	Independent rerun failed without modifying the locked package.

Current status: HANDOFF_READY + INTERNAL_CLEANROOM_PASS

Next action

Send the ZIP to an external reproducer. They should run only the script in 02_REPRODUCTION_SCRIPT and return pass_summary, trial_results, cv_model_losses, Python version, OS, and deviation notes.